

DOMANDA

24

D: 22 E: 24

Dati i segnali

$$x(t) \leftrightarrow X(f)$$

$$y(t) \leftrightarrow Y(f)$$

$$X(f) = \int_{-\infty}^{\infty} dt x(t) e^{-j2\pi ft}$$

$$Y(f) = \int_{-\infty}^{\infty} dt y(t) e^{-j2\pi ft}$$

allora vale $\int_{-\infty}^{\infty} dt x(t) y^*(t) = \int_{-\infty}^{\infty} df X(f) Y^*(f)$

se $x(t) = y(t)$ vale

$$\int_{-\infty}^{\infty} dt |x(t)|^2 = \int_{-\infty}^{\infty} df |X(f)|^2$$

DIMOSTRAZIONE:

è prodotto nel dominio del tempo e la convoluzione nel dominio della frequenza

$$x(t) y^*(t) \leftrightarrow \{X(f)\}^* \{Y^*(-f)\}$$

$$\Rightarrow \{X(f)\}^* \{Y^*(-f)\} \stackrel{\Delta}{=} \int_{-\infty}^{\infty} d\lambda X(\lambda) Y^*(\lambda - f)$$

per la proprietà di invertimento degli assi $Y^*(f - \lambda) = Y^*(\lambda - f)$

Siccome l'area di un segnale e la sua trasformata di Fourier coincidono

$$\int_{-\infty}^{\infty} dt x(t) y(t) = \int_{-\infty}^{\infty} d\lambda X(\lambda) Y^*(\lambda - f) \Big|_{f=0} = \int_{-\infty}^{\infty} d\lambda X(\lambda) Y^*(\lambda) \quad \text{O.k.} \quad \square$$

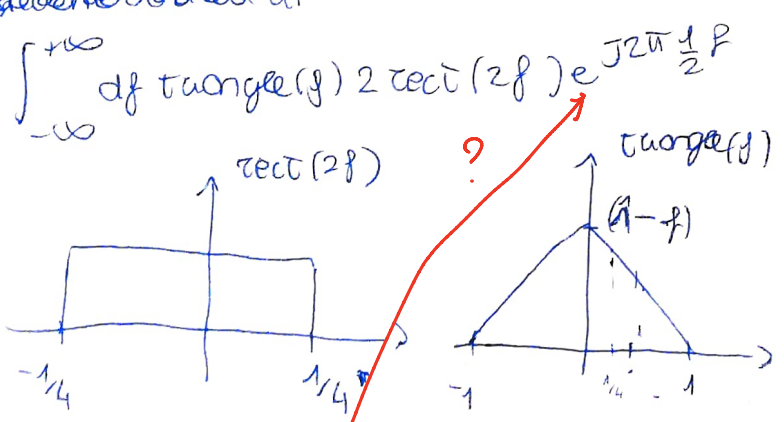
$$\int_{-\infty}^{+\infty} \text{sinc}^2(t) \text{sinc}(-2t-1) dt =$$

considero $x(t) = \text{sinc}^2(t)$ $y(t) = \text{sinc}(-2t-1) = \text{sinc}(t)$, oppure la trasformata di Fourier

$$X(f) = \text{triangle}(f), \quad y(t) = \text{sinc}\left(-\frac{t+1/2}{1/2}\right)$$

$$Y(f) = \frac{1}{2} \text{rect}\left(\frac{f}{1/2}\right) e^{j2\pi \frac{1}{2} f}$$

~~scrittura corretta~~



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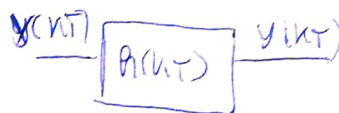


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$$= 2 \int_0^{1/4} \alpha f(1-f) = 2 \left[f - \frac{f^2}{2} \right]_0^{1/4} = 2 \left[\frac{1}{4} - \frac{1}{32} \right] = \frac{7}{16}$$

① $h(nT) = \frac{1}{T} 0,3^{-n} 10(-nT)$ 5/6

$$x(nT) = 1 + \sin(2\pi f_0 nT) \quad , \quad f_0 = \frac{1}{10T}$$



$$x(nT) = 1 + \sin(2\pi f_0 nT) = 1 + \cos(2\pi f_0 nT - \frac{\pi}{2}) = x_1(nT) + x_2(nT)$$

Siccome gli esponenziali complessi sono autofunzioni dei filtri

$$y(nT) = x_1(nT) H(f_0) + \text{exp}(nT) |H(f_0)| x_2(nT + \angle H(f_0))$$

$$H(f) = ?$$

$$\begin{aligned} H(f) &= T \sum_{k=-\infty}^0 \frac{1}{T} 0,3^{-k} e^{-j2\pi f kT} \\ &= \sum_{k=0}^{+\infty} 0,3^k e^{j2\pi f kT} = \sum_{k=0}^{+\infty} (0,3 e^{j2\pi f T})^k \\ &= \frac{1}{1 - 0,3 e^{j2\pi f T}} \end{aligned}$$

$$H(f_0) = H(0) = \frac{1}{1 - 0,3} = \frac{10}{7}$$

$$H(f_0) = H\left(\frac{1}{10T}\right) = \frac{1}{1 - 0,3 e^{j2\pi \frac{1}{10T} T}} = \frac{1}{1 - 0,3 e^{j2\pi \frac{1}{5}}}$$

*
calcolo modulo e fase

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$$② x(t) = \sum_{k=-\infty}^{+\infty} e^{-2t+6k} \quad 1(2t-6k) = \sum_{k=-\infty}^{+\infty} e^{-(2t-6k)} \quad 1(2t-6k)$$

$x(t)$ è periodico de periodo di 3 secondi

4/6

$$x_1(t) = e^{-t} \quad 1(t)$$

$$\tilde{x}_1(nT) = \sum_{k=-\infty}^{+\infty} e^{-(t-\frac{6}{2}k)} \quad 1(-t-\frac{6}{2}k)$$

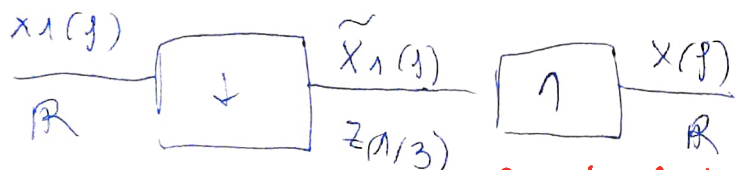
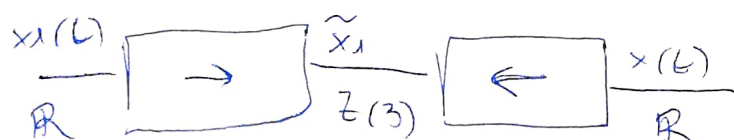
$$X_1(f) = \int_0^{+\infty} e^{-t} e^{-j2\pi ft} dt$$

$$= \int_0^{+\infty} e^{-t(1-j2\pi f)} dt = \left[\frac{e^{-t(1-j2\pi f)}}{-1-j2\pi f} \right]_0^{+\infty} = \frac{1}{1+j2\pi f}$$

$$\tilde{X}_1(f) = \frac{1}{1+j2\pi f} \quad \frac{1}{3}$$

$$X(f) = 3 \sum_{k=-\infty}^{+\infty} \tilde{X}_1(f) \delta_R(t-n3)$$

$$= 3 \sum_n \frac{1}{1+j2\pi f \frac{1}{3}} \delta_R(t-n3)$$



questo fattore?

$$e^{-2t} \quad 1(2t)$$

$$= e^{-2t} \quad 1(t)$$

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$$\dots \text{m}^2 = -12$$

③ $w(nT)$

$$Rw(f) = R_0(1 + \cos(2\pi fT))$$

$$x(nT) = w(nT) + w(nT - T) + w(nT - 2T)$$

$$h(nT) = \delta_z(t)(nT) + \delta_z(t)(nT - T) + \delta_z(t)(nT - 2T)$$

$$R_x = R_w |H(f)|^2$$

$$H(f) = 1 + e^{-j2\pi fT} + e^{-j2\pi f2T} = e^{-j2\pi fT} \left(1 + e^{j2\pi fT} + e^{j2\pi f2T} \right)$$

$$= e^{-j2\pi fT} (1 + 2\cos 2x)$$

$$= e^{-j2\pi fT} (1 + \cos^2 x)$$

$$= e^{-j2\pi fT} (\sin^2 x + \cos^2 x - \cos^2 x) =$$

$$= e^{-j2\pi fT} (1 + 4\cos^2 x - 1) = e^{-j2\pi fT} (4\cos^2 x) ?$$

$$2\cos^2 x = 1 + \cos(2x) ?$$

$$2\cos(2x) = (2\cos^2 x - 1)^2$$

$$R_x = R_0 (1 + \cos(2\pi fT)) |H(f)|^2$$

$$M_x = \int R_x(f) = R_0 \int (1 + \cos(2\pi fT)) |H(f)|^2$$

Incompleto...
Domanda del tempo...

④ $B = 20 \text{ kHz}$

$$M_T = 200 \text{ V}^2$$

$$A_T = 2 \text{ dB/km}$$

$$L = 60 \text{ km}$$

$$R_0 = 10^{-18} \text{ V}^2/\text{Hz}$$

$$\Delta = \frac{M_T A_T^2 m}{2 R_0 B}$$

$$-10 \log_{10} A_T^2 m = A_T \cdot L$$

$$-10 \log_{10} A_T^2 m = 2 \cdot 60 = 120$$

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$$\log_{10} A m^2 = -12$$

$$A m^2 = 10^{-12}$$

$$\Delta = \frac{200 \cdot 10^{-12}}{2 \cdot 10^{-18} \cdot 20k} = 5000$$

$$\textcircled{5} \quad g(t) = v_0 \operatorname{sinc}\left(\frac{2t}{T}\right)$$

$$h(t) = h_0 \operatorname{sinc}^2\left(\frac{t}{T}\right)$$

$$l(t) = l_0 \operatorname{sinc}\left(\frac{6t}{T}\right) \operatorname{sinc}\left(\frac{8t}{T}\right)$$

5/6

$$P_e = \frac{2 \cdot 3 - 2}{2} Q\left(\frac{v_0^2}{20n}\right)$$

$$= 2 Q\left(\frac{v_0^2}{20n}\right)$$

$$G(\beta) = v_0 \frac{T}{2} \operatorname{rect}\left(\beta \frac{T}{2}\right)$$

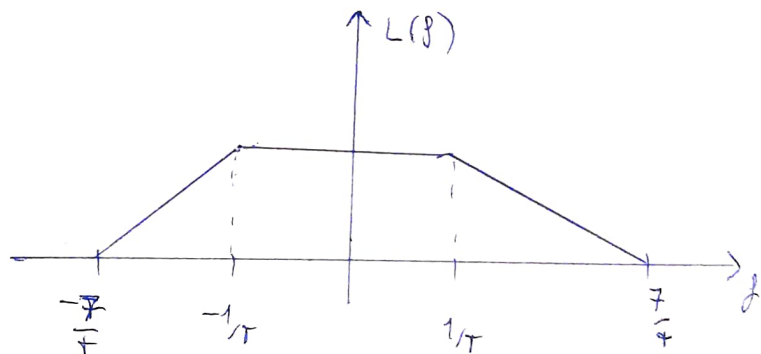
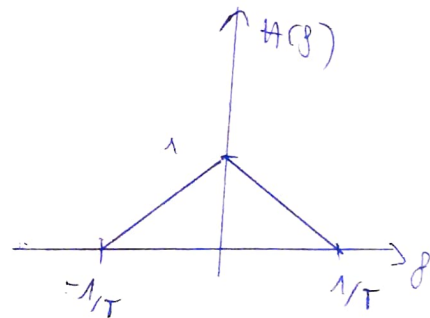
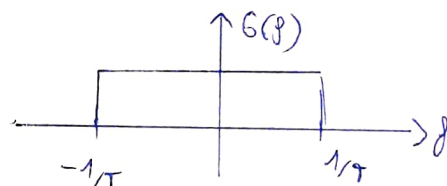
$$H(\beta) = h_0 T \operatorname{triang}\left(\beta T\right)$$

$$L(\beta) = \left\{ l_0 \frac{T}{8} \operatorname{rect}\left(\beta \frac{T}{8}\right) \right\}^* \left\{ \frac{T}{6} \operatorname{rect}\left(\beta \frac{T}{6}\right) \right\}$$

$$\frac{1}{2} \text{ Base maggiore} = \left(\frac{8}{T} + \frac{6}{T}\right) \frac{1}{2} = \frac{7}{T}$$

$$\frac{1}{2} \text{ Base minore} = \left(\frac{8}{T} - \frac{6}{T}\right) \frac{1}{2} = \frac{1}{T}$$

$$\text{altezza} = l_0 \frac{T}{8} \frac{T}{6} \frac{6}{T} = l_0 \frac{T}{8}$$



$$C(\beta) = L(\beta) H(\beta) G(\beta)$$

$$= v_0 \frac{T}{2} h_0 T l_0 \frac{T}{8} \operatorname{triang}(\beta T) \Rightarrow c(t) = v_0 \frac{T}{2} h_0 T l_0 \frac{T}{8} \frac{1}{T} \operatorname{sinc}^2\left(\frac{t}{T}\right)$$

$$C(0) = \frac{v_0}{16} h_0 l_0 T^2 = v_0$$

è di Nyquist.

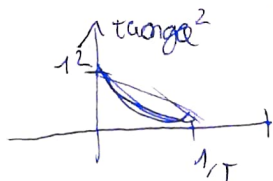
$$R_n = R_0 E h \Rightarrow R_0 \int d\beta |H(\beta)|^2$$

$$M_n = 0 \text{ per } n \neq 0 \Rightarrow m_n = 0$$

$$\Rightarrow \sigma^2 n = R_0 h_0^2 T^2 \int d\beta \operatorname{triang}^2(\beta T)$$

$$= R_0 h_0^2 T^2 \frac{1}{T} \frac{2}{3} = R_0 h_0^2 T \frac{2}{3}$$

$$P_e = \frac{1}{2} Q\left(\frac{v_0^2 h_0^2 l_0^2 T^4}{16^2 R_0 h_0^2 T \frac{2}{3}}\right) = \frac{1}{2} Q\left(\frac{v_0^2}{85n}\right)$$



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